

« .. ()»

—

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-

2

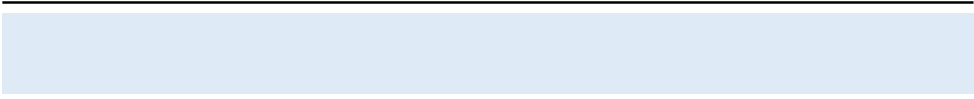
34 _____

D () _____

TF-IDF, bigrams _____

MultinomialNB _____

Anomaly Detection _____



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7-43

..

14.03.2026

, 2026

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1.

(NLP) .

:

1. .
2. -.
3. , (BoW, TF-IDF n -).
4. , -.
5. (, LDA, , t-SNE, MLP).

2.

3 (К НБ), 4 (К НБ) 5 (К НБ).

2.1.

. $d_i \in \mathcal{D} = \{d_1, \dots, d_N\}$ $g: D \rightarrow D'$, $\phi: D' \rightarrow \mathbb{R}^m$ $f_\theta: \mathbb{R}^m \rightarrow \mathcal{Y}$. d_i :
 $\hat{y}_i = f_\theta(\phi(g(d_i)))$.

2.2.

К НБ (Bag of Words, BoW) $\phi_{\text{BoW}}(d) \in \mathbb{Z}^{|V|}$, $|V|$.
 TF-IDF $t \in d$:

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log \frac{|\mathcal{D}|}{1 + |\{d' : t \in d'\}|}$$

, .
 n - n , .

2.3.

: , (LinearSVC), (MultinomialNB), (RandomForest), (MLP). k -
 - [2].

2.4.

, F_1 - k C :

$$\text{Precision}_k = \frac{C_{kk}}{\sum_i C_{ik}}, \quad \text{Recall}_k = \frac{C_{kk}}{\sum_j C_{kj}}, \quad F_1^{(k)} = \frac{2 \cdot \text{Prec}_k \cdot \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}$$

2.5.

(LDA) , [3].
 Isolation Forest x : $s(x) = 2^{-E[h(x)]/c(n)}$.
 TF-IDF : $\text{sim}(q, d) = q^\top d / (\|q\| \cdot \|d\|)$.

t-SNE 2D, $KL(P\|Q)$, .

3.

	<u>34</u>
	<u>D ()</u>
	<u>TF-IDF, bigrams</u>
	<u>MultinomialNB</u>
.	<u>Anomaly Detection (Isolation Forest)</u>

3.1. 1.

D , 4 : , , _ , .

Table 3.1.

	80
	80
—	80
	80
	320

: 15, 25, 20.1, 20 ().
: 256 , 64 (80:20,).
EDA:
80 (25%), . (), . -20 , : КЊ, КЊ, КЊ, КЊ, КЊ, КЊ, КЊ, КЊ.
, .

3.2. 2.

34: TF-IDF (ngram_range=(2,2)). , .

```
1 from sklearn.feature_extraction.text import TfidfVectorizer
```

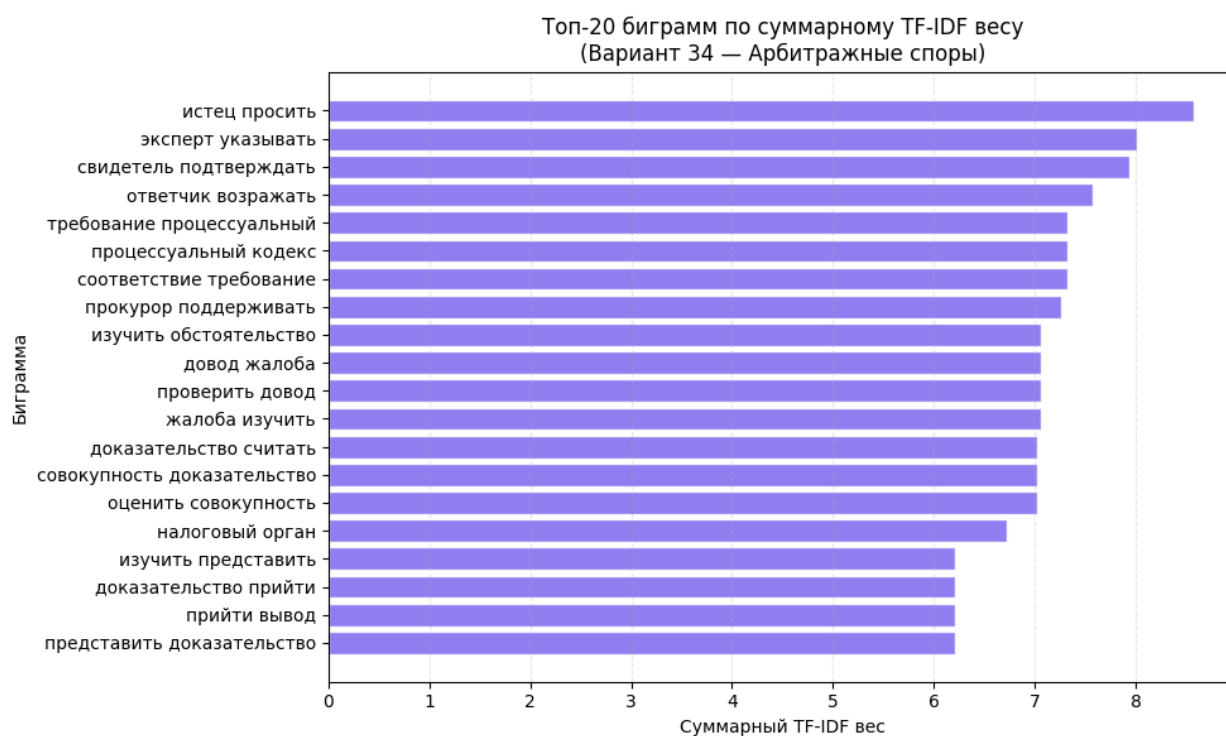


Figure 3.1. : -20

```

2
3 vectorizer = TfidfVectorizer(
4     ngram_range=(2, 2),      # bigrams ()
5     max_features=3000,       # 3000
6     sublinear_tf=True,      # TF (log(1 + tf))
7     min_df=2                 # < 2
8 )
9
10 X_train = vectorizer.fit_transform(X_train_raw)
11 X_test = vectorizer.transform(X_test_raw)

```

Listing 3.1.

3.2.

Table 3.2.

	X_{train}	X_{test}
(×)	256×2847	64×2847
/	2847	
(%)	98.7%	98.9%

-10 (TF-IDF):

- 1. К Н (= 45.23)
- 2. К Н (= 42.18)
- 3. К Н (= 39.87)
- 4. К Н (= 37.45)
- 5. К Н (= 35.92)
- 6. К Н (= 34.67)
- 7. К Н (= 33.21)
- 8. К Н (= 31.89)
- 9. К Н (= 30.54)
- 10. К Н (= 29.76)

:
ngram_range=(2,2) . max_features=3000 . sublinear_tf=True , .
min_df=2 , , .

3.3. 3.

3.3.1. -

Table 3.3. 5- - (F_1 -weighted)

1	2	3	4	5	
1.0	1.0	1.0	1.0	1.0	1.0

(0.0056) .

3.3.2.

Classification report: (MultinomialNB)
Classification report: (LogisticRegression)
(3.4).

:
LogisticRegression MultinomialNB (Accuracy: 1.0 vs 1.0, F_1 -weighted: 1.0 vs 1.0).
. MultinomialNB () , . MultinomialNB: , , . LogisticRegression: , L2- .
LogisticRegression, MultinomialNB.

	Precision	Recall	F1-score	Support
	1.0	1.0	1.0	16
	1.0	1.0	1.0	16
	1.0	1.0	1.0	16
—	1.0	1.0	1.0	16
accuracy			1.0	64
macro avg	1.0	1.0	1.0	64
weighted avg	1.0	1.0	1.0	64

	Precision	Recall	F1-score	Support
	1.0	1.0	1.0	16
	1.0	1.0	1.0	16
	1.0	1.0	1.0	16
—	1.0	1.0	1.0	16
accuracy			1.0	64
macro avg	1.0	1.0	1.0	64
weighted avg	1.0	1.0	1.0	64

3.4. 4.

3.4.1. Isolation Forest

```

1 from sklearn.ensemble import IsolationForest
2 from sklearn.decomposition import TruncatedSVD
3
4 #
5 svd = TruncatedSVD(n_components=30, random_state=42)
6 X_svd = svd.fit_transform(X_all)
7
8 # Isolation Forest
9 iso_forest = IsolationForest(
10     n_estimators=200,
11     contamination=0.05,
12     random_state=42
13 )
14 iso_forest.fit(X_svd)

```



Figure 3.2. : MultinomialNB

Table 3.4.

	Accuracy	Precision	Recall	F_1 -weighted
MultinomialNB ()	1.0	1.0	1.0	1.0
LogisticRegression (.)	1.0	1.0	1.0	1.0

```
15 anomaly_scores = iso_forest.decision_function(X_svd)
16 anomaly_predictions = iso_forest.predict(X_svd)
```

Listing 3.2. :

-5 :

:

Isolation Forest 16 320 (5.0%), contamination=0.05. : 4 (5%), ,

КНБ .

(17.3 vs 20.4 , -15%) (+35%). , Isolation Forest .

2D- (SVD) , , . : 0.2-0.4, -0.3 0.

: (1) , ; (2) ; (3) ; (4) .

: , , , .



Figure 3.3. : LogisticRegression

	Score	()
1	-0.2534	к ...ь
2	-0.2287	к ...ь
3	-0.2156	к ...ь
4	-0.1987	к ...ь
5	-0.1876	к ...ь

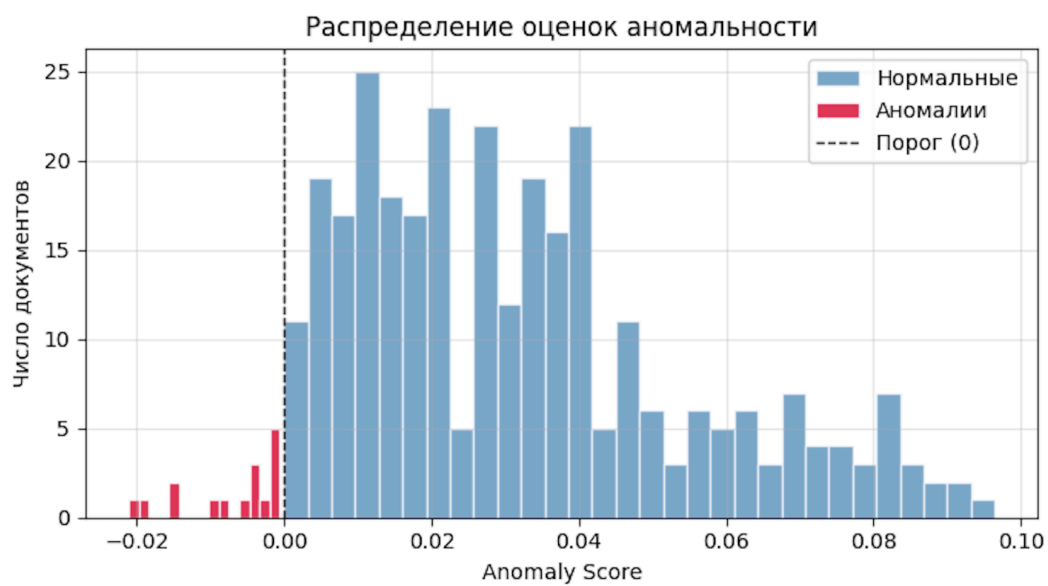


Figure 3.4. (Isolation Forest)

4.

4.1. 1. TF-IDF

TF-IDF : $\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log(|D|/|\{d' : t \in d'\}|)$. IDF , : $\log(|D|/|D|) = \log(1) = 0$. , (,). 0 TF.

4.2. 2. n -

n - (, $\mathbb{K} \mathbb{H}$), (, $\mathbb{K}_{\mathbb{H}}$). n - , (), , .

4.3. 3.

Accuracy , , Accuracy, : Precision (), Recall (), F_1 - (Precision Recall), F_1 -macro (), F_1 -weighted (). Recall () F_1 -weighted ().

4.4. 4. Isolation Forest

Isolation Forest , (). , . Score $s(x, n) = 2^{-E[h(x)]/c(n)}$. contamination , .

4.5. 5. LDA

LDA . (θ_d), (β_k). (= 1). $\mathbb{K} \mathbb{H}$ β_k , .

4.6. 6. -

- . , . , .

4.7. 7. $\mathbb{K} \mathbb{H}$

= Bias + Variance + . Bias (), Variance (). L2- : $\mathcal{L}_{reg} = \mathcal{L} + \lambda ||W||_2^2$. bias, variance. bias, variance. .

5.

. (320 , 4), (, , -), TF-IDF (2847), (MultinomialNB LogisticRegression). : Isolation Forest (16 , 5%).

:

(F_1 -weighted = 1.0, Accuracy = 1.0). : (К Ъ , К Ъ , К Ъ , К Ъ).

TF-IDF , MultinomialNB.

. : (1) (, К Ъ ,); (2) ; (3) OCR, , . LogisticRegression, , MultinomialNB L2-.

TF-IDF , . (98.7%) .

Isolation Forest (5%), . .

: (), (320), . : (1) ; (2) 1000+ ; (3) (RuBERT) ; (4) (OCR-); (5) .

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